

PROJECT REPORT

Objective

The main objective of this project is to develop a Multimodal Emotion Recognition System that predicts human emotions using both speech and text inputs. The system combines Speech Emotion Recognition and Text Emotion Recognition using Deep Learning techniques and fusion learning to improve prediction accuracy and robustness.

The project aims to:

- recognize emotions from speech audio
 - recognize emotions from spoken text
 - combine speech and text features using multimodal fusion
 - improve emotion classification accuracy
 - build a real-time prediction system using Streamlit
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Code Structure

```
project/
|
|— Dataset/
|
|— models/
|  |
|  |— speech_pipeline/
|  |— text_pipeline/
|  └— fusion_pipeline/
|
|— Results/
|
|— deployment/
|  └— app.py
|
|— notebooks/
|  └— main.ipynb
```

|
|— README.md
|— requirements.txt

Architecture Decisions

Speech Pipeline

Architecture Used

- MFCC Feature Extraction
- CNN
- LSTM

Why This Architecture?

Speech contains emotional variations in:

- pitch
- tone
- energy
- speaking style

MFCC extracts meaningful speech features.

CNN helps identify local emotional patterns in speech signals.

LSTM captures temporal dependencies and emotional flow across the audio sequence.

Final Speech Flow

Audio → MFCC → CNN → LSTM → Emotion Prediction

Text Pipeline

Architecture Used

- Tokenization
- Embedding Layer

- LSTM

Why This Architecture?

Text input provides semantic context.

The embedding layer converts words into dense vector representations.

LSTM helps understand contextual and sequential relationships between words.

Final Text Flow

Text → Tokenizer → Embedding → LSTM → Emotion Prediction

Fusion Pipeline

Architecture Used

- Feature Concatenation
- Dense Layers
- Softmax Classifier

Why Fusion?

Speech captures vocal emotion while text captures semantic meaning.

Combining both modalities improves:

- contextual understanding
- prediction robustness
- classification accuracy

Final Fusion Flow

Speech Features

+

Text Features

↓

Fusion Layer

↓

Emotion Prediction

Architecture Summary

Pipeline	Architecture
Speech Pipeline	MFCC + CNN + LSTM
Text Pipeline	Embedding + LSTM
Fusion Pipeline	Concatenation + Dense Layers

The multimodal fusion architecture achieved better performance compared to individual pipelines.

Experiments

Model-Level Experiments

Three different models were developed and evaluated:

- Speech-only model
- Text-only model
- Multimodal fusion model

The performance of all models was compared to analyze the effectiveness of multimodal learning.

Application-Level Experiments

Speech-Only Experiments

Observation

The speech model performed well for emotions with strong vocal variations such as:

- angry
- happy
- sad

Accuracy

Approximately:

- 97%

Limitation

Neutral and pleasant emotions showed overlap due to similar speech characteristics.

Text-Only Experiments

Observation

The text model learned contextual emotional information from spoken words.

Accuracy

Approximately:

- 3%

Limitation

The vocabulary size was limited, reducing contextual diversity.

The same word spoken with different emotions sometimes caused ambiguity.

Multimodal Fusion Experiments

Observation

The fusion model achieved the best overall performance.

Accuracy

Approximately:

- 98%

Why Fusion Performed Better

Speech provides vocal emotional cues while text provides contextual meaning.

Combining both reduced ambiguity and improved classification performance.

Overall Experimental Findings

Model	Accuracy
Speech-only	97%
Text-only	3%
Multimodal Fusion	98%

The fusion model consistently outperformed unimodal models.

Analysis

Which Emotions Were Easiest and Hardest to Classify?

Easiest Emotions

Angry

Angry emotion was easier to classify because:

- speech intensity is high
 - pitch variation is strong
 - vocal energy is distinctive
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Happy

Happy emotion showed clear expressive vocal patterns and energetic speech characteristics.

Hardest Emotions

Neutral

Neutral emotion was difficult because:

- emotional variation is minimal
- speech characteristics overlap with sad and pleasant emotions

Pleasant

Pleasant emotion sometimes overlapped with happy emotion due to similar positive speech patterns.

When Does Fusion Help Most?

Fusion helped most when:

- **speech alone was ambiguous**
- **text alone lacked emotional information**
- **emotions had overlapping vocal patterns**

Fusion improved contextual understanding by combining:

- **speech tone**
 - **spoken content**
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Error Analysis

Failure Case 1

Actual Emotion

Pleasant

Predicted Emotion

Happy

Reason

Both emotions share similar positive vocal expressions.

Failure Case 2

Actual Emotion

Neutral

Predicted Emotion

Sad

Reason

Low-energy speech patterns appeared similar.

Failure Case 3

Actual Emotion

Fear

Predicted Emotion

Surprise

Reason

Both emotions contain sudden pitch changes.

Failure Case 4

Actual Emotion

Disgust

Predicted Emotion

Angry

Reason

Harsh vocal tone similarities caused confusion.

Failure Case 5

Actual Emotion

Happy

Predicted Emotion

Pleasant

Reason

Semantic and vocal overlap between positive emotions.

Emotion Cluster Separability Analysis

Temporal Modelling Block (Speech)

Speech embeddings showed clearer clusters for:

- **angry**
- **happy**
- **sad**

Neutral and pleasant emotions showed overlap because their vocal characteristics are less distinctive.

Contextual Modelling Block (Text)

Text clusters were less separable because:

- **vocabulary size is small**
- **same words appear across multiple emotions**

Text alone was insufficient for strong emotional separation.

Fusion Block

Fusion embeddings showed:

- **improved cluster separation**
- **reduced overlap**
- **clearer emotional boundaries**

Combining speech and text improved discriminative learning.

Cluster Visualization Summary

Block	Cluster Quality
Temporal Modelling	Moderate to High
Contextual Modelling	Moderate
Fusion Block	High

The fusion block produced the most distinguishable emotion clusters.

Overall Analysis

The multimodal fusion model demonstrated superior performance compared to speech-only and text-only systems.

Speech features captured emotional tone while text features provided semantic context.

Fusion learning improved:

- accuracy
 - robustness
 - emotion separability
 - overall prediction quality
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Results and Discussion

Overall Performance

The proposed multimodal system successfully recognized emotions using combined speech and text inputs.

The fusion model achieved the highest classification accuracy.

Speech Performance

The speech model effectively captured vocal emotional variations using:

- MFCC features

- CNN
- LSTM

Strong emotions such as angry and happy were recognized accurately.

Text Performance

The text model captured contextual emotional information using:

- embeddings
- sequential learning

Performance was limited by vocabulary size.

Fusion Performance

Fusion significantly improved:

- emotion classification
- contextual understanding
- robustness

The multimodal system reduced ambiguity between similar emotions.

Key Findings

- Fusion learning outperformed unimodal systems
 - Speech provided strong emotional cues
 - Text improved contextual understanding
 - Combining modalities reduced prediction errors
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Discussion Summary

The proposed architecture successfully demonstrated the effectiveness of multimodal deep learning for emotion recognition.

The combination of speech and text improved classification performance and produced more reliable emotion predictions.

Conclusion

This project successfully developed a Multimodal Emotion Recognition System using speech and text fusion.

Deep learning architectures including CNN and LSTM were used for speech and text modelling.

The fusion model achieved better accuracy compared to individual pipelines by combining vocal emotional information and contextual textual meaning.

The Streamlit deployment provided a user-friendly interface for real-time emotion prediction.

The project demonstrated the effectiveness of multimodal fusion learning in improving emotion recognition performance.

Future Work

Use Emotionally Rich Text Datasets

Larger and more expressive text datasets can improve contextual learning and text emotion classification.

Increase Speaker Diversity

Training on diverse speakers can improve:

- generalization
 - robustness
 - real-world applicability
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Advanced Fusion Techniques

Future systems may use:

- **attention mechanisms**
- **transformer-based fusion**
- **multimodal transformers**

to further improve performance.

Real-Time Deployment

The project can be extended into:

- **live microphone emotion recognition**
- **cloud deployment**
- **real-time streaming systems**
- **mobile applications**